IndicDoc Verify

- **Project title:** IndicDoc Verify: Deep-Learning Layout Analysis for Pre-Print Checking of Indian Question Papers and Their Translations
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- St. Francis Institute of Technology (SFIT)

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- **Academic year / semester:** 2026–27, Semester V (TE AIML)

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Literature Survey Summary Table

Ref	Author	Problem	Dataset	Method	Limitation
[1]	Nath et al., 2025	Layout parsing of Indian-language pages	IndicDLP: 119,806 pages, 12 languages, 42 labels incl. questions and options	Benchmark of YOLOv10x, DocLayout-YOLO, RoDLA, DINO, DiT, Florence-2	Detects regions only; nobody uses the question labels to check papers
[2]	Kasuba et al., 2025	Page-level handwritten Indic text recognition	CHIPS, ~25K pages, 10 languages	DBNet word detection + 6 recognisers (CRNN, PARSeq, ...)	User must name the language; text only
[3]	Mondal et al., 2026	Multilingual table detection	UniTabBank: 55K pages, 82K tables, 28 languages	Semi-automatic labelling; YOLOv11 detector	Borderless tables and degraded scans fail
[4]	Pal et al., 2024	Table question answering, low-resource Indic	BanglaTabQA 2M, HindiTabQA 643K	SQL-generated QA; fine-tuned mBART / M2M / Llama	Wikipedia tables only
[5]	Xu et al., 2021	Multilingual document understanding	30M docs; XFUND, 7 languages	LayoutXLM: text + layout + image transformer	No Indian language
[6]	Kasuba et al., 2025	Grounding answers on the page	MGVG: 70 circulars, 509 QA pairs	OCR + Llama-3.1-8B + region matching	OCR-dependent; single page
[7]	Bhattacharyya et al., 2025	Hindi OCR with vision-language models	1.1M printed Hindi words + pages	LoRA-tuned GOT-OCR 2.0	Hindi only; mixed-language lines are hard
[8]	Dumada & Shah, 2026	Devanagari table extraction	500+ Devanagari documents	Structure + Tesseract content, weighted fusion	No ablation; self-reported
[9]	Babayeva et al., 2025	Translating a test into other languages with an LLM	Force Concept Inventory, many languages	GPT-4o translation + back-translation + human review	Every item still checked by hand; no page structure
[10]	Rei et al., 2022	Translation quality estimation without a reference	WMT 2022 QE shared task	CometKiwi: multilingual encoder predicts quality per sentence	Plain sentences: no question numbers, options or marks

Research Gaps

#	Gap in existing work	How this project addresses it
G1	Exam papers are translated into up to 13 languages and vetted only by hand; errors reach candidates	An automatic pre-print check of every language version against the source
G2	Translation-quality tools [9, 10] read plain sentences: blind to question numbers, options, marks, figures	Compare versions question by question using the page layout itself
G3	Indic layout models [1] detect questions and options, but this is not used to check papers	Use the detector's question / option regions (AP@50 0.81–0.90) as the backbone of the checker
G4	Indic OCR misreads Latin digits in mixed text [7]; a checker of numbers needs every digit right	A "number pass": two OCR models must agree on every number

- Court e-filing defect detection already exists (Supreme Court with IIT Madras; a commercial tool reported in 16 High Courts), so we chose the problem **nobody automates yet**: checking a question paper and its translations before printing.

Problem Statement and Objectives

- **Problem:** a question paper is set in one language and translated into Hindi and regional languages; every version is checked by hand. When a number, option, mark or question differs, lakhs of candidates are affected (grace marks, re-exams, court cases).
- **Main objective:** an offline deep-learning system that reads a printed paper in any of 12 Indian languages and flags every structural or numerical error, and every difference from its source version, before printing.
- **Specific objectives:**
  - Detect the paper's structure (questions, options, answers) with a layout detector; compare 3 detectors.
  - Read text with reliable numbers in 12 languages (number pass).
  - Checks: numbering, duplicates, options, marks, translation parity, repeats, format profiles.
  - Measure errors found and false alarms on a benchmark; ship a web app and API in Docker.

- **Expected output:** a verdict (ready / check / do not print) and every finding with its question and page, outlined on the page, as HTML and JSON.

It has already happened (news reports)

Exam	What went wrong
NEET-UG 2018	49 questions wrong in the Tamil version; HC ordered 196 grace marks, SC set it aside
KPSC KAS 2024	59 errors in the Kannada paper; re-exam for ~2 lakh candidates, errors again
APPSC Group-I	51 of 120 questions alleged mistranslated into Telugu; SC notice
UGC-NET Jun 2026	Translation errors + questions repeated from old papers; re-test 9–10 Sep
MBBS (NTRUHS) 2026	The same MCQ printed twice; AP HC ordered a compensatory mark

Proposed System, Methodology and Novelty

- **Proposed solution:** IndicDoc Verify reads the paper as it will be printed, finds its structure with a deep-learning layout detector, reads the text, and checks it alone and against its source version.
- **Workflow (input to output):**

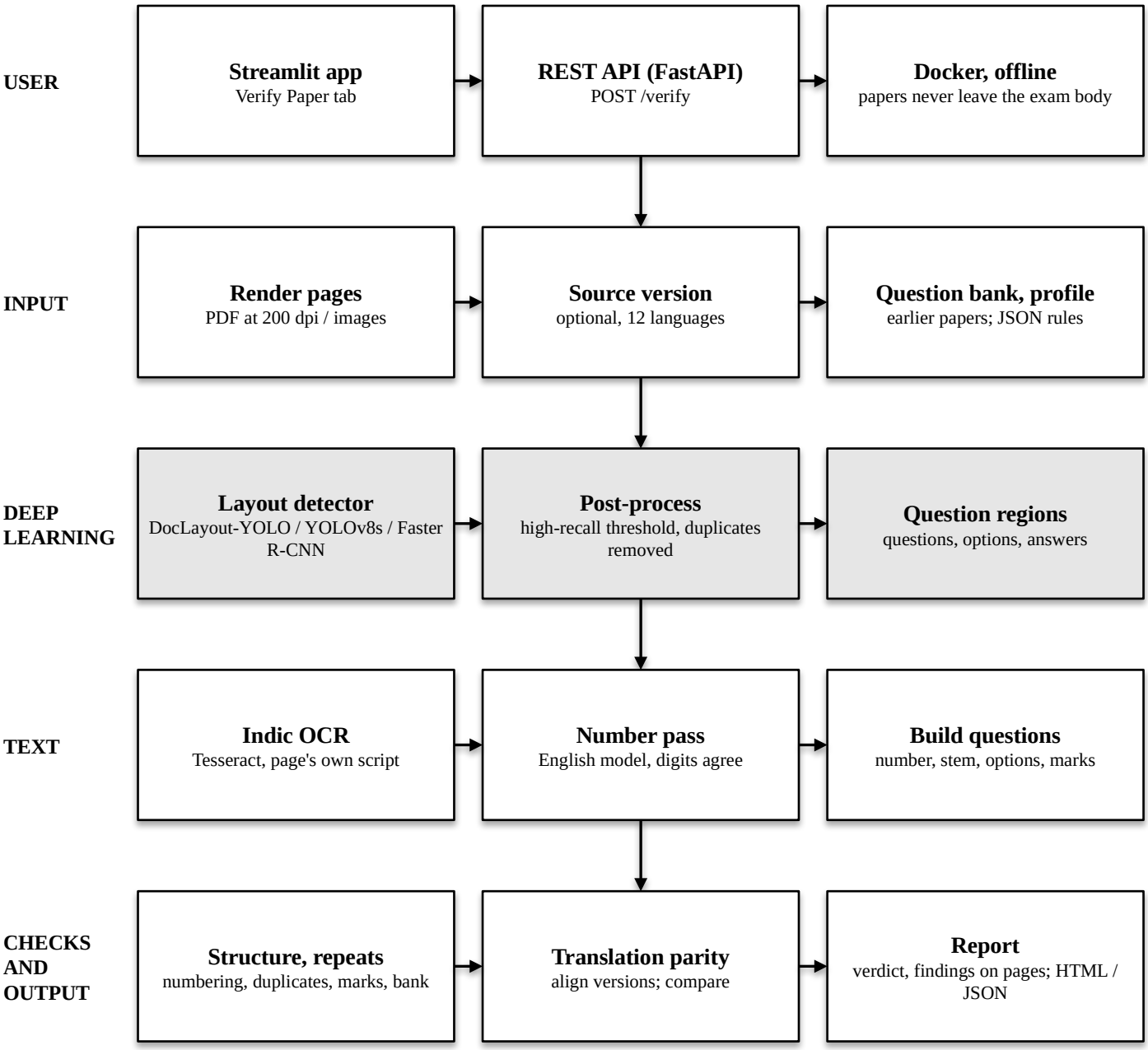
1. **Paper in:** PDF / scans + source version
2. **Find structure:** DocLayout-YOLO: questions, options
3. **Read text:** Indic OCR + English number pass
4. **Build questions:** number, stem, options, marks, figures
5. **Check:** structure, parity, repeats, format
6. **Report:** verdict + findings on the page

- **How deep learning is used:** a convolutional object detector (DocLayout-YOLO) finds every question, sub-question, option and answer on the page (AP@50 0.81–0.90 on these labels); its regions turn a page image into structured questions in any script.

- **What is different (novelty):**
  - **Translation parity without translation:** numbers, option counts, marks, sub-questions and figures must match in every language, so versions are compared question by question with no MT model.
  - **Number pass:** two OCR models must agree on every digit.
  - **USP:** no tool we found checks an Indian question paper against its translations question by question; this runs offline, in 12 languages, before printing.

System Block Diagram

- Major components and the flow of data through the system (top to bottom).

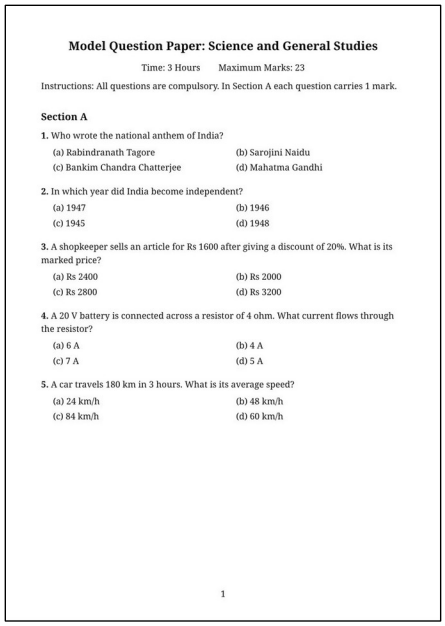


Grey: deep-learning stage. The source version takes the same path as the paper.

Dataset Description and Pre-processing

- **Dataset name and source:** IndicDLP (Nath et al., ICDAR 2025), Hugging Face ai4bharat/indicdlp, AIKosh; 119,806 pages, 12 languages, 12 domains, 42 layout labels.
- **Samples and classes (our subset):** 3,317 pages, 49,995 boxes, 42 classes; 195 question-paper pages, 1,227 question and 1,209 option boxes.
- **Split (official):** train 2,293 / validation 342 / test 682 pages.
- **Verification benchmark:** 132 English–Hindi and English–Tamil paper pairs (18 questions each), 108 with one injected error of 9 real kinds, 24 clean; half degraded like scans. A separate development set (48 pairs) was used to build the parser.
- **Pre-processing:** Parquet → PNG + COCO JSON, boxes clipped, COCO → YOLO labels, letterbox 640 / 1024 px; PDF pages rendered at 200 dpi.
- **Augmentation:** YOLOv8 mosaic, HSV, scale; Faster R-CNN colour jitter, blur; no mirror flips.
- **Challenges:** class imbalance ("paragraph" ≈ 32% of boxes); no public set of erroneous exam papers, hence the benchmark.

Actual data used



Model Question Paper: Science and General Studies

Time: 3 Hours Maximum Marks: 23

Instructions: All questions are compulsory. In Section A each question carries 1 mark.

Section A

1. Who wrote the national anthem of India?

(a) Rabindranath Tagore (b) Sarojini Naidu
(c) Bankim Chandra Chatterjee (d) Mahatma Gandhi

2. In which year did India become independent?

(a) 1947 (b) 1946
(c) 1945 (d) 1948

3. A shopkeeper sells an article for Rs 1600 after giving a discount of 20%. What is its marked price?

(a) Rs 2400 (b) Rs 2000
(c) Rs 2800 (d) Rs 3200

4. A 20 V battery is connected across a resistor of 4 ohm. What current flows through the resistor?

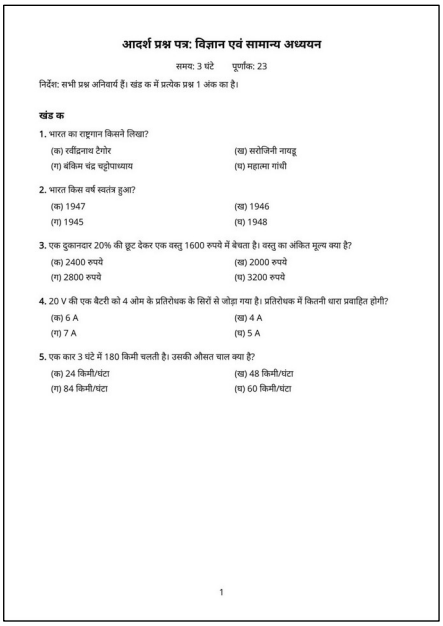
(a) 6 A (b) 4 A
(c) 7 A (d) 5 A

5. A car travels 180 km in 3 hours. What is its average speed?

(a) 24 km/h (b) 48 km/h
(c) 84 km/h (d) 60 km/h

1

Benchmark: English source



आदर्श प्रश्न पत्र: विज्ञान एवं सामान्य अध्ययन

समय: 3 घंटे पूर्णांक: 23

निर्देश: सभी प्रश्न अनिवार्य हैं। खंड क में प्रत्येक प्रश्न 1 अंक का है।

खंड क

1. भारत का राष्ट्रगान किसने लिखा?

(क) रवींद्रनाथ टैगोर (ख) सरोजिनी नायडू
(ग) बंकिम चंद्र चट्टोपाध्याय (घ) महात्मा गांधी

2. भारत किस वर्ष स्वतंत्र हुआ?

(क) 1947 (ख) 1946
(ग) 1945 (घ) 1948

3. एक दुकानदार 20% की छूट देकर एक वस्तु 1600 रुपये में बेचता है। वस्तु का अंकित मूल्य क्या है?

(क) 2400 रुपये (ख) 2000 रुपये
(ग) 2800 रुपये (घ) 3200 रुपये

4. 20 V की एक बैटरी को 4 ओम के प्रतिरोधक के सिरों से जोड़ा गया है। प्रतिरोधक में कितनी धारा प्रवाहित होगी?

(क) 6 A (ख) 4 A
(ग) 7 A (घ) 5 A

5. एक कार 3 घंटे में 180 किमी चलती है। उसकी औसत चाल क्या है?

(क) 24 किमी/घंटा (ख) 48 किमी/घंटा
(ग) 84 किमी/घंटा (घ) 60 किमी/घंटा

1

Benchmark: Hindi version



केसीआर ने गरीबों के लिए काम नहीं किया

'आप' विधायक अमानतुल्लाह के परिसरों पर ईडी के छापे

आठ्यमएफ ने भारत की वृद्धि दर का अनुमान बढ़ाकर 6.3 फीसद किया

दिल्ली की सभा से पहले सिखों मामले में सुप्रीम कोर्ट ने गुजरात सरकार से मुफ्त माफी की मांग पर निर्णय लेने में कितना समय लगता है?

विशेषाधिकार समिति की बैठक में नहीं पहुंचे सांसद रमेश बिधूड़ी

राजस्थान में महिला निशानेबाजों से बलात्कार व छेड़छाड़, प्रशिक्षक के खिलाफ प्राथमिकी

संजय सिंह को ईडी हिरासत 13 अक्तूबर तक बढ़ी

इजराइल ने गाजा में हमले तेज किए, तीन लाख अतिरिक्त सैनिक तैनात

रोजेसवा में टिकट नहीं मिलने पर भाजपा में फूटे बागी सुर

आर्थिक जनगणना के लिए केंद्र सरकार को करेंगे मजबूर : राहुल

अदेंपति के खिलाफ मुकदमे की मांगी

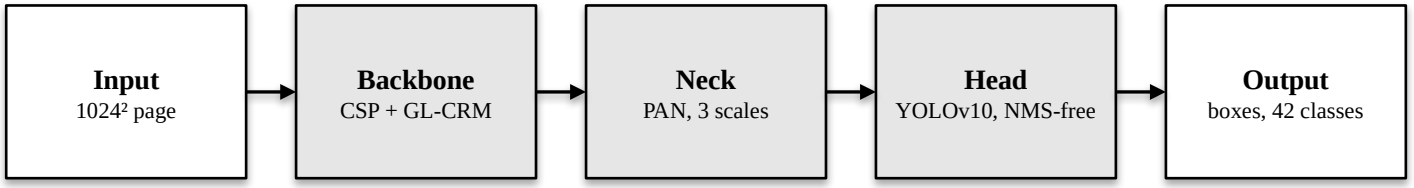
संकट की घड़ी में इजराइल के साथ खड़ा है भारत

IndicDLP: Hindi newspaper

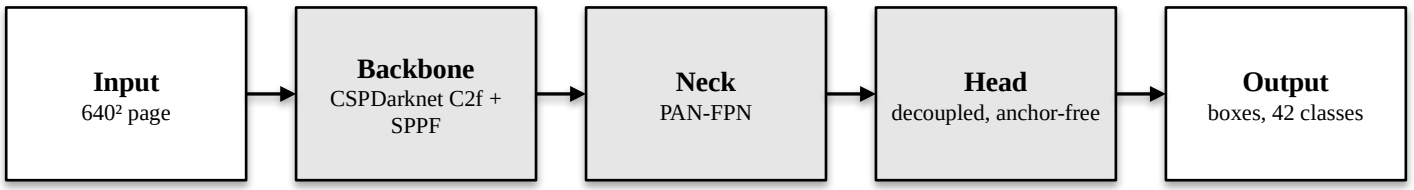
Deep Learning Model Architecture

- **Model selected:** DocLayout-YOLO (20.0 M parameters), fine-tuned on the full IndicDLP by its authors (pretrained, transfer learning); compared with YOLOv8s and Faster R-CNN fine-tuned by us.
- **Input / output:** a page image (1024×1024×3 for DocLayout-YOLO, 640×640×3 for YOLOv8s) → N boxes, each with one of 42 labels and a score.
- **Modifications:** COCO heads replaced by 42-class heads (43 with background for Faster R-CNN); losses CIoU + DFL + BCE (YOLO), RPN + CE + smooth-L1 (Faster R-CNN).

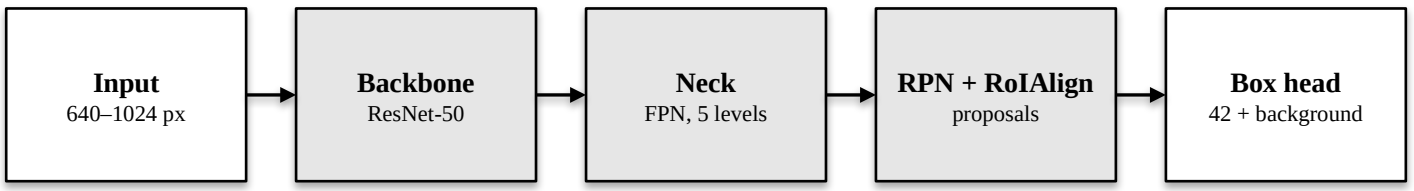
DocLayout-YOLO, 20.0 M (used by IndicDoc Verify)



YOLOv8s, 11.2 M (fine-tuned by us)



Faster R-CNN, 41.5 M (baseline)



Grey: learned feature extraction and detection layers.

Tools, Technologies and Environment

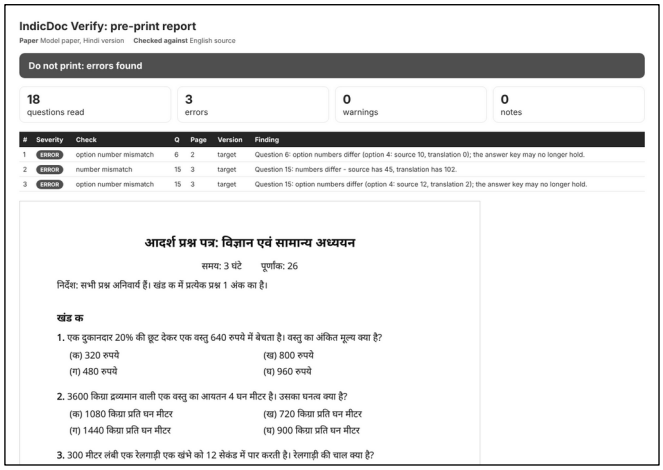
Component	Details to report
Programming language	Python 3.11 (ML, checks, app, API); JavaScript (Node) for the benchmark renderer
DL framework	PyTorch 2 with torchvision; Ultralytics 8 (YOLOv8); doclayout-yolo 0.0.4
Development environment	VS Code / terminal, Git + GitHub, pytest (unit tests for every check)
Libraries	pycocotools, OpenCV, Pillow, NumPy, pandas, Matplotlib, PyMuPDF; Tesseract 5 (tessdata_best, 12 scripts); tesseract.js for the benchmark
Hardware	Apple M1 Pro GPU via PyTorch MPS (training, evaluation); CPU container for the benchmark; CPU in Docker
Dataset tools	Hugging Face Hub + fsspec (partial Parquet reads); COCO / YOLO converters; Playwright + Noto fonts to render benchmark papers
Interface / deployment	Streamlit app (Verify Paper tab) + FastAPI (/verify); Docker + docker-compose, offline

Implementation Details

- **What we implemented:** IndicDoc Verify (ml/verify), the OCR stage with the number pass (ml/ocr), training and evaluation of two detectors, the benchmark, a Streamlit app and a REST API in Docker.
- **Data loading:** IndicDLP Parquet → COCO; PDFs rendered at 200 dpi (PyMuPDF).
- **Model:** pretrained DocLayout-YOLO at its high-recall threshold (0.15); YOLOv8s and Faster R-CNN fine-tuned from COCO weights.
- **Loss / optimiser:** YOLOv8s CIoU + DFL + BCE, auto optimiser; Faster R-CNN SGD, lr 0.005, batch 4.
- **Inference:** layout → OCR per region → question building → checks → report.
- **Checks:** numbering, duplicates, options, marks; translation parity by Needleman–Wunsch alignment; repeats by character 3-gram similarity; JSON format profiles.
- **Interface:** Streamlit "Verify Paper" tab and POST /verify (JSON or HTML report).

```
# every page: layout (DL) -> OCR with number pass
out = predict_single(model, page, "doclayout", device, conf=0.15)
doc = extract_document(page, out, "hin", number_pass=True)

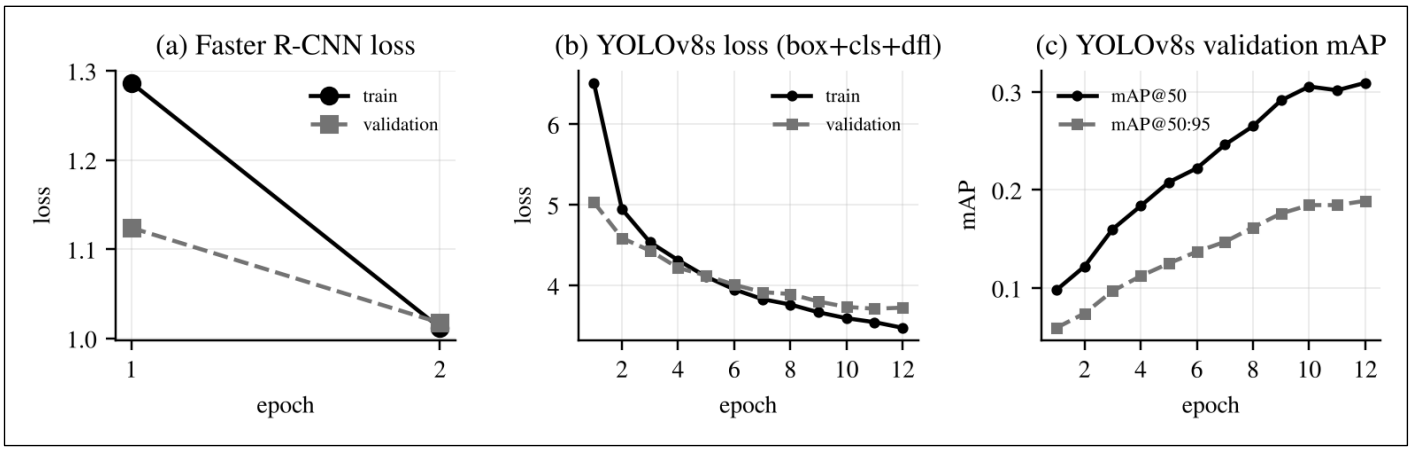
# ml/verify: questions, checks, parity, repeats
report = verify(hindi_pages, source_pages=english_pages,
                bank=earlier_questions, profile="question_paper")
html = to_html(report, {"paper": page_images})
```



Verification report produced by the system

Experimental Setup and Training

- **Dataset sizes:** train 2,293 / validation 342 / test 682 pages; benchmark 132 paper pairs (test) + 48 (development).
- **Configuration:** Faster R-CNN batch 4, SGD 0.005, **2 epochs, 76 min**; YOLOv8s 640 px, batch 8, **12 epochs, 60 min**; Apple M1 Pro GPU (PyTorch MPS).
- **E1 Layout:** 3 detectors on 682 test pages: mAP@50, mAP@50:95, P / R / F1, per label.
- **E2 Robustness:** blur, noise, JPEG, half resolution, faded print.
- **E3 Verification:** injected errors, real Tesseract OCR; an error counts only if found at the right question.
- **E4 Ablations:** without the number pass; without layout-based options; with a noisy layout.
- **Protocol:** parser built on the development set; test set run once.



Training and validation curves of the two detectors we trained (measured).

Results and Performance Evaluation

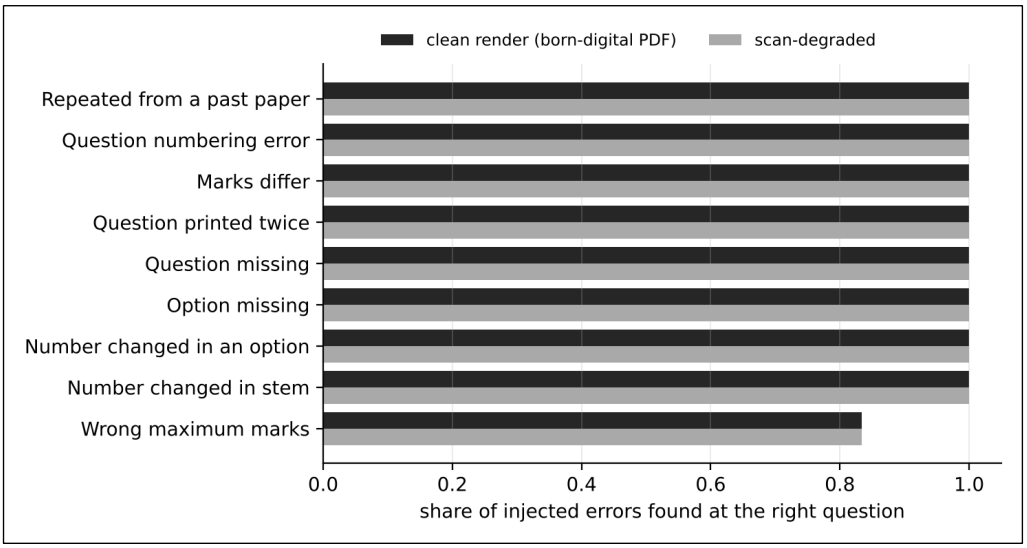
Layout detection (682 test pages)

Model	mAP@50	mAP@50:95	Precision	Recall	F1
DocLayout-YOLO (authors)	0.759	0.575	0.848	0.775	0.810
YOLOv8s (ours)	0.259	0.165	0.603	0.508	0.551
Faster R-CNN (ours)	0.148	0.079	0.510	0.435	0.469

AP@50 on question-paper labels

Label	DocLayout-YOLO	YOLOv8s	Faster R-CNN
first-level-question	0.854	0.415	0.240
second-level-question	0.898	0.045	0.018
third-level-question	0.872	0.026	0.000
options	0.839	0.553	0.243
answer	0.809	0.000	0.000

- **Verification:** 98% of injected errors found at the right question on born-digital papers, 98% on scan-degraded pages.
- **False alarms:** 3.0 per clean pair (born-digital), 5.5 (degraded); each is shown on the page for the vetter.
- **Robust:** 98% still found with a noisy layout; 132/132 papers had every question recovered.



Errors found at the right question, per error type.

Comparison with Baseline / Existing Methods

Layout detectors (metric: COCO mAP)

Detector	Training data	mAP@50	mAP@50:95
Faster R-CNN (ours, baseline)	2,293 pages, 76 min	0.148	0.079
YOLOv8s (ours)	2,293 pages, 60 min	0.259	0.165
DocLayout-YOLO (authors)	~96K pages	0.759	0.575
Published DocLayout-YOLO [1]	full test set	≈0.735	0.545

Verification ablation (test set)

System	Errors found	False alarms / clean pair
Full system	98%	4.5
Without number pass	77%	13.8
Without layout-based options	98%	11.5
Noisy layout	98%	7.5

Existing approaches

Approach	Checks the paper?	Compares versions?	Indic, offline?
Manual vetting (status quo)	yes, by hand	by hand	yes
Answer-sheet AI (Eklavvya, GradeLab)	no, grades answers	no	partly
MT quality estimation (CometKiwi [10])	no structure	sentences only	cloud / GPU
LLM back-translation [9]	no structure	with human review	cloud
IndicDoc Verify (ours)	**numbering, options, marks, repeats**	**question by question**	**12 languages, offline**

- YOLOv8s beats the Faster R-CNN baseline (0.259 vs 0.148 mAP@50) at 9× the speed; the released full-data model is best and is used.
- The number pass is the largest single gain: without it, errors found fall to 77% and false alarms triple.

Sample Outputs and Demonstration

14. 10 संख्याओं का औसत 52 है। यदि संख्या 68 हटा दी जाए, तो शेष संख्याओं का औसत क्या होगा?

(क) 80 (ख) 70

(ग) 30 (घ) 50

15. 6 संख्याओं का औसत 33 है। यदि संख्या 102 हटा दी जाए, तो शेष संख्याओं का औसत क्या होगा?

(क) 48 (ख) 42

(ग) 30 (घ) 12

खंड ख

16. ओम का नियम लिखिए और उसका सूत्र दीजिए। [3 अंक]

17. भारतीय संविधान की तीन विशेषताएँ लिखिए। [5 अंक]

18. प्रकाश संश्लेषण की प्रक्रिया समझाइए। [3 अंक]

Correct: *a changed number in the Hindi version found and outlined on the page.*

3. भारत का संविधान किस वर्ष अंगीकृत किया गया था?

(क) 1947 (ख) 1948

(ग) 1950 (घ) 1949

4. समुद्र तल पर शुद्ध जल किस तापमान पर उबलता है?

(क) 90 °C (ख) 100 °C

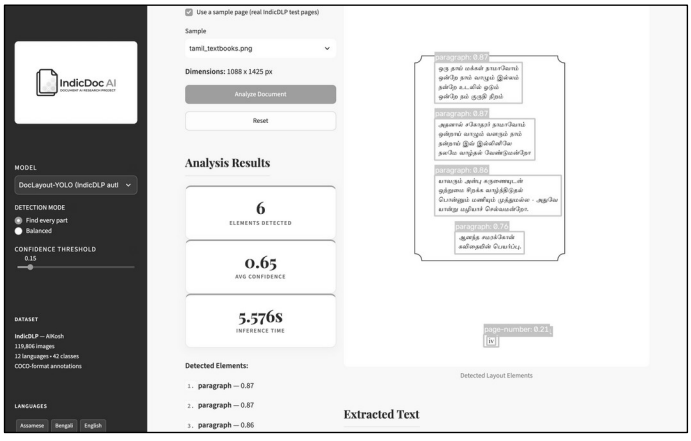
(ग) 110 °C (घ) 120 °C

5. 60 V की एक बैटरी को 12 ओम के प्रतिरोधक के सिरों से जोड़ा गया है। प्रतिरोधक में कितनी धारा प्रवाहित होगी?

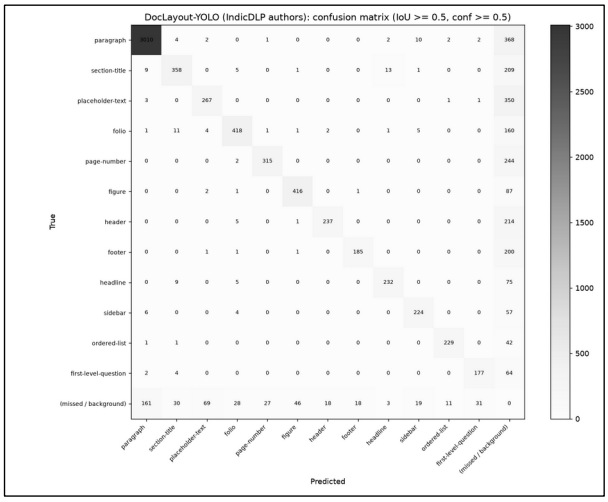
(क) 3 A (ख) 8 A

(ग) 4 A (घ) 5 A

Incorrect: *a false alarm on a degraded scan (OCR lost a digit of an option).*



Running application: layout regions of a Tamil page



Confusion matrix of the layout model

Challenges and Limitations

- **OCR is the weakest link:** Indic Tesseract models misread Latin digits ("1947" → "947") and glue option markers to values; the number pass, layout-based option splitting and page-level digit-script detection were needed to bring false alarms down.
- **No public data of real errors:** Exam bodies do not publish erroneous pre-print papers, so we built a benchmark with realistic injected errors, kept a separate development set, and report clean and scanned results separately.
- **Meaning is not checked yet:** Parity catches numbers, options, marks and missing questions without translation; a pure change of meaning needs a cross-lingual model (hook present, not yet evaluated).
- **Compute and data size:** Our detectors saw 2.3K pages for about an hour (0.26 vs 0.76 mAP@50 for the released model), so the product uses the released model; our contribution is the checker built on it.

Conclusion and Future Scope

- **Conclusion**
  - **Implemented:** IndicDoc Verify: layout detection (3 detectors compared), OCR for 12 scripts with a number pass, question parsing, structure / parity / repeat / format checks, report, web app, REST API, Docker.
  - **Results:** 98% of injected errors found at the right question on born-digital papers (98% on degraded scans), 3.0 false alarms per clean pair; best layout model 0.759 mAP@50.
  - **Objectives:** all four met; our own detectors did not reach the released model, which we report openly.

- **Future scope**
  - Pilot with a university exam cell or a state board on real pre-print papers.
  - Meaning check with a cross-lingual sentence model; number words and formulas.
  - Better Indic OCR behind the same interface; answer-key consistency check.
  - More document profiles: bilingual government circulars (Official Languages Act s.3(3)), forms, certificates.
  - Deployment: offline licence per exam body (NTA, UPSC, state PSCs, boards, universities).

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- **Other sources:**
  - Dataset: huggingface.co/datasets/ai4bharat/indicdlp (AIKosh)
  - Weights: zenodo.org/records/15881917
  - Tesseract OCR 5 (tessdata_best), tesseract.js 7
  - Cases: The News Minute (NEET 2018), Deccan Herald / ETV Bharat (KPSC 2024-25), India Legal (APPSC), The Week / SCC Online (UGC-NET 2026), Medical Dialogues (MBBS 2026)
  - Code: github.com/va-ada/IndicDoc-AI